

VPC CHALLENGE

Use Case: Setting up Transit Gateway and VPC Endpoints for a Multi-VPC Architecture

Scenario:

A large organization is migrating its on-premises infrastructure to the AWS cloud.

The organization's architecture involves multiple VPCs for different departments and applications, each requiring secure communication with centralized services and external resources.

The IT team needs to design and implement a scalable and efficient network architecture to accommodate the organization's growth and ensure robust connectivity between VPCs and external services.

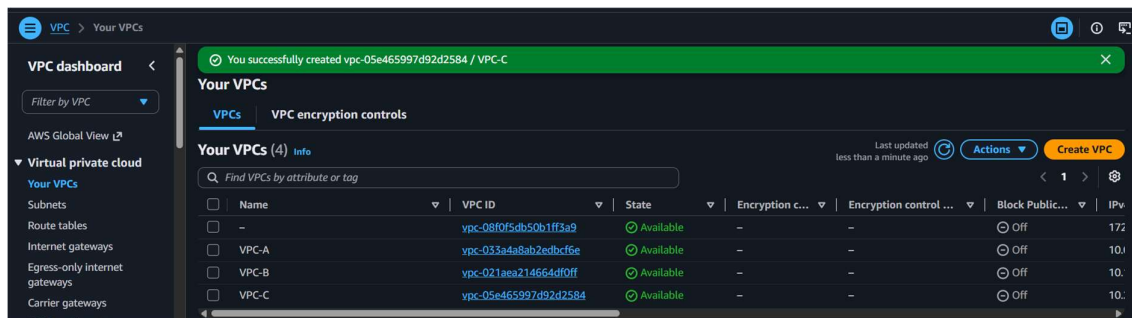
Objectives:

- **Design and deploy a scalable network architecture using AWS Transit Gateway to simplify network connectivity between multiple VPCs.**
- **Configure VPC endpoints to securely access AWS services without internet gateways or NAT gateways, ensuring data privacy and minimizing exposure to external threats.**

Step 1: Create VPCs

Create three VPCs.

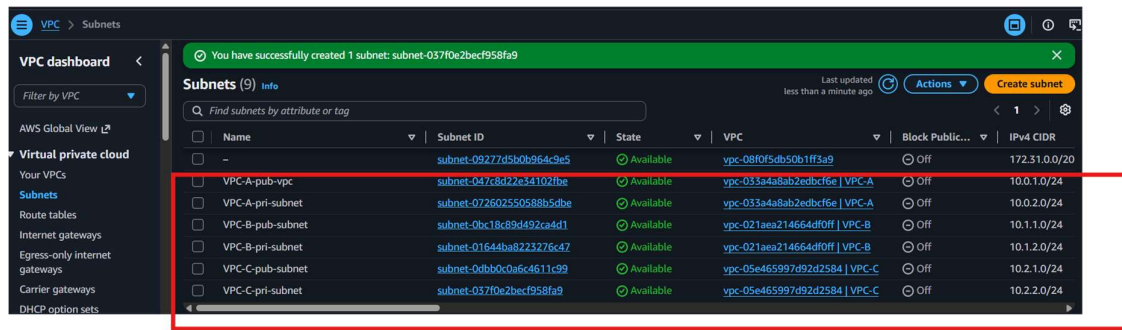
- VPC-A (10.0.0.0/16)
- VPC-B (10.1.0.0/16)
- VPC-C (10.2.0.0/16)



Create:

- 1 Public Subnet
- 1 Private Subnet

- Updated route table and Internet gateway

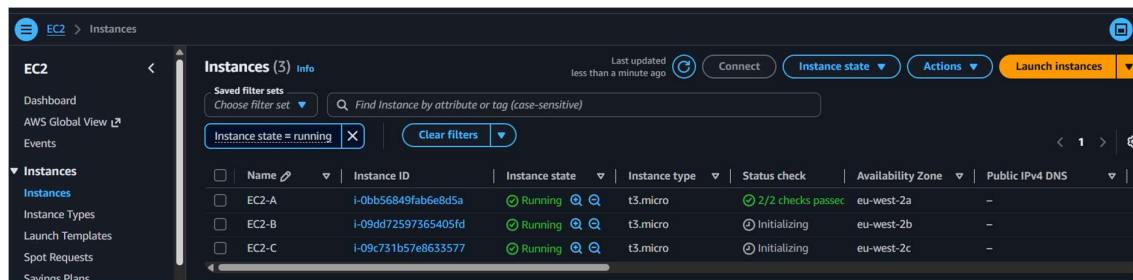


Step 2: Launch EC2 Instances

Launch one EC2 instance in each VPC.

Example:

- EC2-A
- EC2-B
- EC2-C



Step 3: Create Transit Gateway

Go to

VPC → Transit Gateways → Create Transit Gateway

Provide

- Name
- Amazon ASN (Default)
- Enable DNS Support

Create transit gateway [Info](#)

A transit gateway (TGW) is a network transit hub that interconnects attachments (VPCs and VPNs) within the same AWS account or across AWS accounts.

Details - optional

Name tag
Creates a tag with the key set to Name and the value set to the specified string.

vpc-tgw

Description [Info](#)
Set the description of your transit gateway to help you identify it in the future.

vpc-tgw

Configure the transit gateway

Amazon side Autonomous System Number (ASN) [Info](#)

4200000000

Click **Create**.

VPC > Transit gateways

Site-to-Site VPN connections
Client VPN endpoints
AWS Verified Access instances
Verified Access trust providers
Verified Access groups
Verified Access endpoints

You successfully created tgw-0c589f0d318fb12c2 / vpc-tgw.

You can visualize and monitor your Transit Gateway(s) from the AWS Network Manager. Register your Transit Gateway by creating a global network to get started.

Transit gateways (1) [Info](#)

Find transit gateway by attribute or tag

☐	Name	Transit gateway ID	Owner ID	State
☐	vpc-tgw	tgw-0c589f0d318fb12c2	265243686408	Pending

Actions **Create transit gateway**

Step 4: Attach VPCs to Transit Gateway

Go to

Transit Gateway → Attachments → Create Attachment

Attach:

- VPC-A

VPC > Transit gateway attachments > Create transit gateway attachment

Create transit gateway attachment [Info](#)

A transit gateway (TGW) is a network transit hub that interconnects attachments (VPCs and VPNs) within the same AWS account or across AWS accounts.

Details

Name tag - optional
Creates a tag with the key set to Name and the value set to the specified string.

VPC-A-tga

Transit gateway ID [Info](#)

tgw-0c589f0d318fb12c2

Attachment type [Info](#)

VPC

- VPC-B

VPC > Transit gateway attachments > Create transit gateway attachment

Create transit gateway attachment Info

A transit gateway (TGW) is a network transit hub that interconnects attachments (VPCs and VPNs) within the same AWS account or across AWS accounts.

Details

Name tag - optional
Creates a tag with the key set to Name and the value set to the specified string.

VPC-B-tga

Transit gateway ID Info

tgw-0c589f0d318fb12c2

Attachment type Info

VPC

- VPC-C

Transit gateway attachment: tgw-attach-0bc4725d95bae26cb / VPC-C-tga

Details | Flow logs | Tags

Details

Transit gateway attachment ID	State	Resource type	Association state
tgw-attach-0bc4725d95bae26cb	Pending	VPC	-
Transit gateway ID	Resource owner ID	Resource ID	Association route table ID
tgw-0c589f0d318fb12c2	265243686408	vpc-05e465997d92d2584	-

Select the subnet in each VPC.

Wait until the attachment status becomes **Available**.

VPC > Transit gateway attachments

Transit gateway attachments (3) Info

Find transit gateway attachment by attribute or tag

☐	Name	Transit gateway attachment ID	Transit gateway ID	State	Resource type	Resource ID
☐	VPC-A-tga	tgw-attach-0b44eadf860d8bb9c	tgw-0c589f0d318fb12c2	Available	VPC	vpc-033a4a8ab2edbcf6e
☐	VPC-B-tga	tgw-attach-03a3f1d5a9d428355	tgw-0c589f0d318fb12c2	Available	VPC	vpc-021aea214664df0ff
☐	VPC-C-tga	tgw-attach-0bc4725d95bae26cb	tgw-0c589f0d318fb12c2	Available	VPC	vpc-05e465997d92d2584

Step 5: Update Route Tables

In every VPC Route Table, add routes:

For VPC-A

Destination Target

10.1.0.0/16 Transit Gateway

10.2.0.0/16 Transit Gateway

VPC

Route tables

rtb-0d0c0a0a0c0e0a0e

Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	Active	No	CreateRoute
10.1.0.0/16	igw-0da29205b986903b	-	No	CreateRoute
10.2.0.0/16	tgw-0c589f0d318fb12c2	-	No	CreateRoute

For VPC-B

10.0.0.0/16 Transit Gateway

10.2.0.0/16 Transit Gateway

VPC

Route tables

rtb-0213c7197c11e7570

Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.1.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	Active	No	CreateRoute
10.0.0.0/16	igw-094a6a382a84db8a5	-	No	CreateRoute
10.2.0.0/16	tgw-0c589f0d318fb12c2	-	No	CreateRoute

For VPC-C

10.0.0.0/16 Transit Gateway

10.1.0.0/16 Transit Gateway

VPC

Route tables

rtb-07bde704d2393f1ff

Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin	
10.2.0.0/16	local	Active	No	CreateRouteTable	
Q 0.0.0.0/0	Internet Gateway	Active	No	CreateRoute	Remove
Q 10.0.0.0/16	igw-0350832240d4d86d5	-	No	CreateRoute	Remove
Q 10.1.0.0/16	Transit Gateway	-	No	CreateRoute	Remove
Q 10.1.0.0/16	tgw-0c589f0d318fb12c2	-	No	CreateRoute	Remove

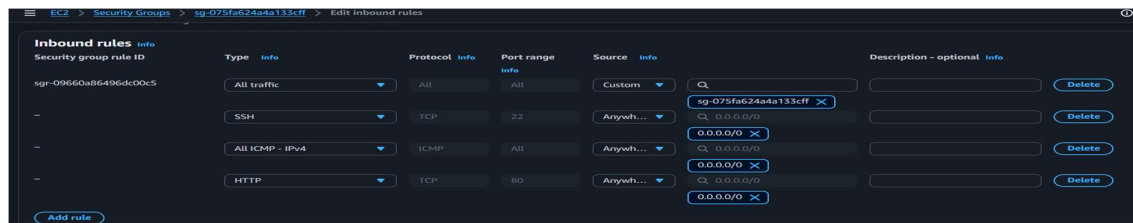
Step 6: Configure Security Groups

Allow ICMP (Ping)

or

Allow required ports like:

- SSH (22)
- HTTP (80)
- HTTPS (443)



Step 7: Test Connectivity

From EC2-A

```
ping <Private-IP-of-EC2-B>
```

```
ping <Private-IP-of-EC2-C>
```

Connectivity should succeed through the Transit Gateway.

```
[ec2-user@ip-10-0-1-160 ~]$ ping 10.1.1.23
PING 10.1.1.23 (10.1.1.23) 56(84) bytes of data.
64 bytes from 10.1.1.23: icmp_seq=469 ttl=126 time=7.19 ms
64 bytes from 10.1.1.23: icmp_seq=470 ttl=126 time=0.929 ms
64 bytes from 10.1.1.23: icmp_seq=471 ttl=126 time=0.977 ms
64 bytes from 10.1.1.23: icmp_seq=472 ttl=126 time=0.949 ms
64 bytes from 10.1.1.23: icmp_seq=473 ttl=126 time=0.923 ms
64 bytes from 10.1.1.23: icmp_seq=474 ttl=126 time=0.944 ms
64 bytes from 10.1.1.23: icmp_seq=475 ttl=126 time=0.965 ms
64 bytes from 10.1.1.23: icmp_seq=476 ttl=126 time=0.953 ms
64 bytes from 10.1.1.23: icmp_seq=477 ttl=126 time=0.977 ms
64 bytes from 10.1.1.23: icmp_seq=478 ttl=126 time=0.923 ms
64 bytes from 10.1.1.23: icmp_seq=479 ttl=126 time=0.943 ms
64 bytes from 10.1.1.23: icmp_seq=480 ttl=126 time=0.956 ms
64 bytes from 10.1.1.23: icmp_seq=481 ttl=126 time=0.935 ms
64 bytes from 10.1.1.23: icmp_seq=482 ttl=126 time=0.932 ms
64 bytes from 10.1.1.23: icmp_seq=483 ttl=126 time=0.948 ms
```

Step 8: Create VPC Endpoint

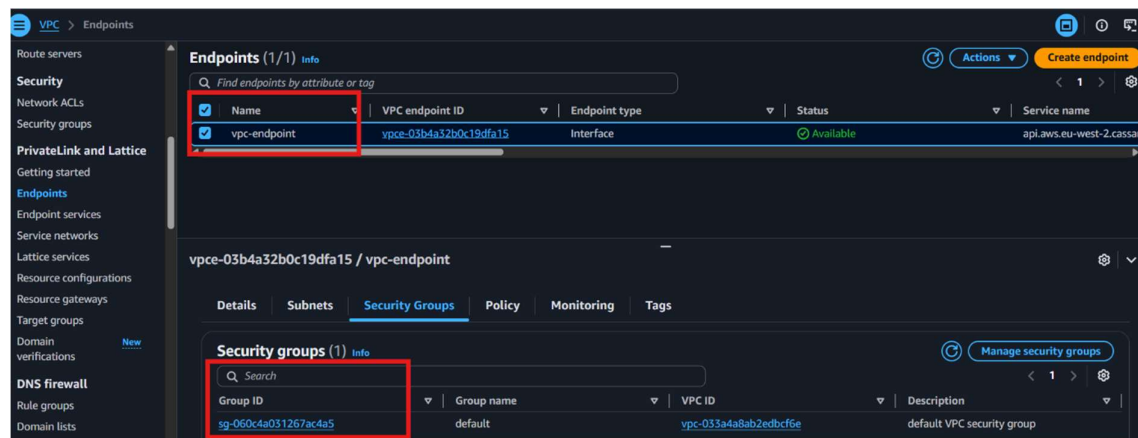
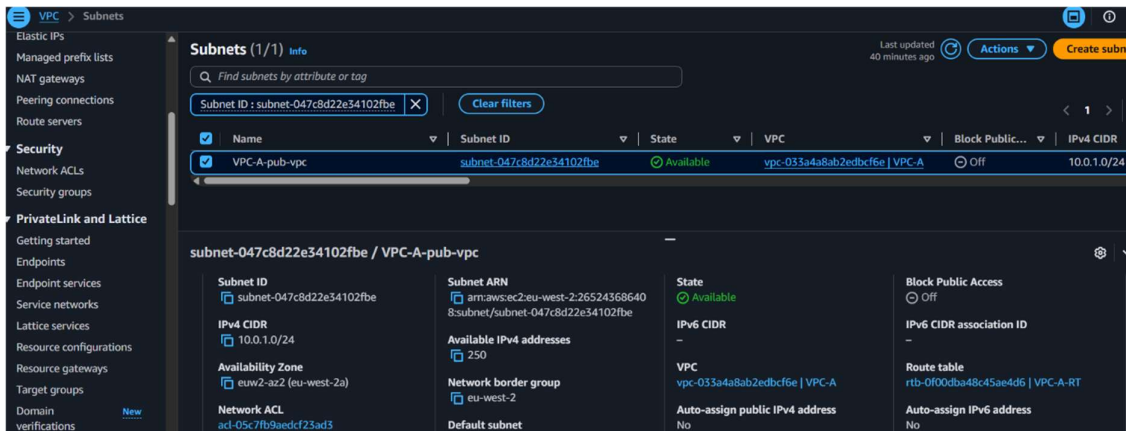
VPC → Endpoints → Create Endpoint

Choose the AWS service.

Example:

- S3 (Gateway Endpoint)
- DynamoDB (Gateway Endpoint)

- SSM (Interface Endpoint)



Configure VPC endpoints to securely access AWS services without internet gateways or NAT gateways.

An S3 Gateway Endpoint is specifically designed so EC2 instances can access Amazon S3 without using an Internet Gateway or NAT Gateway.